

Book Buddy

Project Documentation

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1. Overview

Book Buddy is a reading tracker built for the Sayone Tech freshers hiring task. It covers the core requirement of tracking books and reading progress, plus the optional AI features, using the Gemini API to auto fill book details, generate recommendations, and write short reviews from user notes.

The app is a two part setup. The React frontend talks to a FastAPI backend directly, and the backend handles the database and the calls to Gemini.

2. Features

Core

- Add books with title, author, genre, and status (wishlist, reading, completed)
- Track progress by logging pages read instead of moving a slider
- Book marks itself completed once logged pages reach the total page count
- Notes and a rating out of 5 for completed books
- Delete books

AI features (Gemini)

- Leave author, genre, or page count blank when adding a book, Gemini fills them in from the title
- AI recommendations, three suggested titles based on books marked reading or completed
- AI generated review, turns notes and a rating into a short review

Engagement

- Reading streak, counts consecutive days with at least one logged reading session
- Badge for books finished in the current calendar year
- Reading goal progress bar

3. Data Model

Book

Field	Purpose
title, author, genre	Basic book info, author and genre may be filled by AI
status	wishlist, reading, or completed
total_pages, current_page	Used for the progress bar and percent complete
notes, rating	Filled in once a book is completed
added_date, completed_date	Used for the yearly badge

Reading Log

Field	Purpose
book_id	Which book the session belongs to
pages_read	Pages logged in that session
log_date	Used to calculate the reading streak

4. Key Logic

- Adding a book with missing fields sends one Gemini prompt asking for whatever is missing, read back as simple lines rather than JSON, since smaller models sometimes wrap JSON in extra text
- Logging pages that reach the total page count automatically marks the book completed
- The streak counts backward day by day from today through logged dates, and stops at the first gap
- Recommendations need at least two books marked reading or completed

5. API Reference

Method	Endpoint	Description
POST	/books	Add a book

GET	/books	List books
PATCH	/books/{id}	Update status, notes, rating, progress
DELETE	/books/{id}	Delete a book
POST	/books/{id}/log-pages	Log pages read
POST	/books/{id}/ai-review	Generate an AI review
GET	/recommend	Get AI recommendations
GET	/stats	Completion stats
GET	/streak	Current reading streak
GET	/badges	Books finished this year

6. Trade-offs

- Chose page based logging over a plain percent slider, more natural for readers and it enables auto completion and the streak feature
- Skipped ISBN lookup and real login to protect the deadline
- Used plain CSS instead of a component library, kept things light for this size of project

7. Known Limitations and Future Work

- No rate limiting on the AI endpoints
- No automated tests yet, everything has been tested manually
- No caching on AI recommendation calls, identical requests hit Gemini again each time
- Would move to Postgres if this needed to support more concurrent use